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LevUp

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About the Company

BeyondIRR Group operates across three synergistic business lines that together create a comprehensive financial ecosystem. Its InvestEdge platform delivers technology product solutions to banks, wealth firms, and independent financial advisors, with a strong client base that includes one of the top three private banks, one of the top three wealth firms, a leading broking firm, and one of the oldest and largest wealth firms in the country. Through its Private Markets vertical, BeyondIRR focuses on deals and investment banking in unlisted companies, aggregating demand and supply for pre-IPO and unlisted shares via partnerships with leading wealth businesses, while also securing access to exclusive, hard-to-reach investment opportunities for wealth firms and family offices. Complementing this, LevUp, its wealth and asset management arm, offers quant-driven investment strategies powered by algorithmic models to deliver data-backed, systematic investment solutions.

Objective

Design a fully systematic, rules-based, long-only equity strategy using Indian equity market data. You will receive historical daily data from 2010–2020 to research and develop your strategy.

Your final Python code must be frozen and submitted after testing on 2010–2020 data. Once submissions close, you will receive unseen data for 2020–2025 in the exact same format. You must run the same code without any modification on this unseen dataset and submit the performance report.

Final evaluation will be done only on the 2020–2025 results, combined with assessment of your signal logic and robustness.

Dataset Format (Both Given & Unseen)

You will receive data in the below format:
date, symbol, open, high, low, close, value, volume, sector, mcap, lms, in_nse500

Where:

Column	Description
date	Trading date (YYYY-MM-DD)
symbol	Stock identifier
open	Opening price
high	Intraday high
low	Intraday low
close	Closing price
value	Shares traded
volume	Opening price
sector	Sector classification
mcap	Market capitalization (₹)
lms	Market-cap bucket (Large / Mid / Small)
in_nse500	1 if the stock was part of NSE 500 that day, else 0

Important:

- Only daily OHLC and provided variables will be available.
- IPOs appear only from their listing date.
- Delisted stocks disappear from the dataset after delisting.
- No other external datasets may be used.
- You may derive unlimited calculated variables from the given data.

You'll also receive price data for major indices.

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Portfolio & Capital Rules

Your strategy must satisfy:

Rule	Requirement
Portfolio type	Long-only
Initial capital	₹50,00,000 (50 lakhs)
Fractional shares	Not allowed
Minimum stocks held	1
Maximum stocks held	100
Rebalance frequency	≥ 1 day (you choose the interval))
Execution	T+1 using (O+H+L+C)/4 plus transaction costs if buying T+1 using (O+H+L+C)/4 - transaction costs if selling
Cash allocation	Allowed up to 100% cash
Full investment	Also permitted

Transaction Costs (Per Transaction)

Applied automatically on each buy and sell:

Component	Rate
STT	0.10%
GST on STT	18% of STT
Brokerage (including GST)	0.10%
Slippage	0.05%

Cost=STT 0.10%+ GST 0.018%+ Brokerage 0.10%+ Slippage 0.05%
≈ 0.268% per side ≈ 0.536% round-trip

Your code must apply these costs at the execution price.

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Execution Assumption (Fixed & Mandatory)

1. Signal generated using data up to day T
2. Trade executed on day T+1
3. Execution Price = Average(Open, High, Low, Close of T+1)
4. Then transaction costs are deducted
5. No same-day execution allowed.
6. Avoid look-ahead bias.

Strategy Logic Rules

Your strategy must be:
✓ Fully systematic, deterministic, and rule-based
✓ Implemented only using Python
✓ Capable of running unchanged on unseen data
✓ Based only on supplied dataset columns

You may:
✓ Construct unlimited custom indicators using the given data
✓ Use machine learning — but you must disclose your features and ensure no forward-looking leakage

You may not:
× Use external datasets
× Hard-code future events
× Manually adjust holdings
× Fit parameters using unseen data

Performance Reporting Requirements

For both 2010–2020 and 2020–2025, your code must compute:
Core Metrics
- CAGR
- Annualized Standard Deviation of daily returns
- Maximum Drawdown
- Information Ratio vs Nifty 500

- Up-capture vs Nifty 500
- Down-capture vs Nifty 500

Rolling Performance Metrics
Calculate rolling outperformance vs Nifty 500:

Window	Required Outputs
1 Year	Average Outperformance & Worst Underperformance
3 Year	Average Outperformance & Worst Underperformance
5 Year	Average Outperformance & Worst Underperformance

Rolling windows should be overlapping and calculated daily.

Scoring Framework

Final scoring will consist of:

A. Strategy Construction & Signal Logic — 30% Weight
Judged on:

- Concept clarity
- Feature selection
- Portfolio construction logic
- Risk controls
- Robustness
- Avoidance of obvious bias / curve-fitting
- Interpretability

B. Out-of-Sample Strategy Performance — 70% Weight
Based on a weighted score (equal weight to each) derived from:

- CAGR
- Volatility
- Sharpe characteristics via risk-return metrics
- Max Drawdown
- Information Ratio vs Nifty 500
- Up/Down Capture
- Rolling outperformance & worst underperformance statistics

Only 2020–2025 performance counts toward ranking. Each participant will be normalized (minmax) on all the above parameters (equal weight) and cumulative score shall be the average of the normalized score.

Bias, Robustness & Disqualification

Submissions may be disqualified for:

- Forward-looking bias
- Using future information
- Overfitting via excessive tuning
- Hard-coding Nifty returns or future events
- Curve-fit rules detected in audit
- Code failing to run
- Unrealistic execution assumptions
- Ignoring transaction costs
- Excessive parameter mining
- Violation of any other rule of Kriti 2026

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Code Requirements

Requirment	Rule
Language	Python only
Libraries	Any allowed
Re-runnability	Must run end-to-end
Data input	Must accept any dataset in same schema
ML models	Allowed but must disclose features & avoid leakage

ML approaches must document:

- Input features
- Lookback windows
- Feature construction method

Black-box entries with unclear logic may be penalized.

Model Execution & Evaluation Flow

Phase 1 — Development (2010–2020 Data)

You will:

- Build & tune your strategy
- Generate performance report
- Submit:
  - Python code
  - Documentation PDF
  - 2010–2020 performance report

Phase 2 — Submission Lock

No further edits allowed.

Phase 3 — Unseen Data Release (2020–2025)

You will:

- Run the same code without modification
- Generate the unseen-period report
- Submit results

Organizers may independently run your code for verification.

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Capital & Position Handling Rules

- No leverage
- No short selling
- No fractional shares
- Minimum 1 stock (except cash calls)
- Maximum 100 stocks

Corporate actions are assumed already reflected in price history.

Output Expectations

Your Python code must output:

- NAV / equity curve
- Drawdown curve
- All required statistics
- Rolling outperformance tables
- Portfolio turnover
- Position counts over time
- Trade log (preferred but optional)